

Physics Engine & Prediction Report

Part 3 + Part 4 - Corrosion RC Beam Optimizer

Generated: April 15, 2026 - 15:21

Equation approach: DIRECT

Equation: $-0.0045141946 \cdot RI^{0.4337962} \cdot (CSI^{0.70042914} - b) \cdot (d + 2 \cdot fc - 192.42197) + 4.026101 \cdot \rho_t - 1.402699$

Data: 804 specimens

Critical corrosion level: $\eta^* = 0.5\%$

Stage A: Symbolic Calculus

Equation (DIRECT): $-0.0045141946 \cdot R_I^{**0.4337962} \cdot (C_S I^{**0.70042914} - b) \cdot (d + 2 \cdot f_c - 192.42197) + 4.026101 \cdot \rho_{ho_t}$

$$df/d_eta = 0$$

$$d2f/d_eta2 = 0$$

$$d3f/d_eta3 = 0$$

$$df/d_d = -0.0045141946 \cdot R_I^{**0.4337962} \cdot (C_S I^{**0.70042914} - b)$$

$$df/d_b = 0.0045141946 \cdot R_I^{**0.4337962} \cdot (d + 2 \cdot f_c - 192.42197)$$

$$df/d_fc = -0.0090283892 \cdot R_I^{**0.4337962} \cdot (C_S I^{**0.70042914} - b)$$

$$df/d_rho_t = 4.02610100000000$$

Critical Point: $\eta^* = 0.50\%$ (method: spline)

Taylor (3rd order): 13.2158787920666

Stage B: Non-Dimensionalization

Poly2: $R^2 = 0.3546$

ExpDecay: $R^2 = -0.0$

Pade: $R^2 = 0.353$

Best: Poly2 ($R^2=0.3546$)

Compound parameter: $\omega^{0.56} * (1-\eta/100)^{0.54}$ | $R^2=0.5666$

Stage C: Sensitivity & Phase Diagram

Sobol First-Order (S1):

CSI: 0.0226

RI: 0.0796

b: 0.2284

d: 0.4751

fc: 0.0693

rho_t: 0.0063

Sobol Total-Order (ST):

CSI: 0.0289

RI: 0.1158

b: 0.3126

d: 0.5674

fc: 0.0848

rho_t: 0.0064

Spider Sensitivity:

CSI: 0.0494

RI: 0.2113

b: 0.5463

d: 1.6409

fc: 0.7489

rho_t: 0.1875

Stage D: Prediction & Validation

Predicted collapse: $\eta = 90\%$

Regime boundaries:

safe_limit: 0.2%

warning_limit: 0.5%

critical_limit: 0.8%

Degradation rates:

$\eta=5\%$: 0.0 per %

$\eta=10\%$: 0.0 per %

$\eta=15\%$: 0.0 per %

$\eta=20\%$: 0.0 per %

$\eta=30\%$: 0.0 per %

$\eta=40\%$: 0.0 per %

$\eta=50\%$: 0.0 per %

Stage E: Physical Validation

Checks passed: 2 / 5

[FAIL] eta=0 (Mmax should be $\sim M_{ACI}$)

predicted_kNm: 13.22

expected_ACI_median_kNm: 10.78

deviation_%: 22.55

[FAIL] eta=100 (Mmax should be ~ 0)

predicted_kNm: 13.22

[PASS] monotonicity (f decreases with eta)

monotonic_%: 100.0

violations: 0

[PASS] positivity (f ≥ 0 for eta in [0,64])

min_value: 13.2159

n_negative: 0

[FAIL] dimensional_consistency

note: DIRECT approach: equation has units of kN.m. Dimensional consistency depends on

[FAIL] equation_vs_all_data

R2: 0.9392

RMSE_kNm: 5.95

n_samples: 804

Stage F: Dimensionless Dataset (Buckingham Pi)

A fully dimensionless dataset has been prepared using Buckingham Pi theorem. This dataset can be used to re-run PySR on Pi-groups directly, producing a universal scaling law valid in ANY unit system (the AI Feynman / Science Advances approach).

File: /kaggle/working/corrosion-rc-beam-optimizer/final_results/physics/dimensionless_dataset_for_pysr.csv

Samples: 804

Columns: Pi_M, Pi_omega, Pi_geom, Pi_bar, Pi_eta, Pi_rho, Pi_M_aci, Pi_R

Target: Pi_R (correction ratio) or Pi_M (dimensionless moment)

Correlation with Pi_M:

Pi_M_aci: 0.7643

Pi_omega: 0.7179

Pi_rho: 0.5866

Pi_bar: 0.3286

Pi_geom: 0.0698

Pi_R: -0.0293

Pi_eta: -0.3971

Stage G: Testable Predictions for Lab Validation

Beam specification: b=150.0mm, d=250.0mm, fy=400.0MPa, fc=30.0MPa, rho=1.5%

Total scenarios: 13 (3 extrapolations)

File: /kaggle/working/corrosion-rc-beam-optimizer/final_results/physics/testable_predictions_for_lab.csv

eta_m (%)	Mmax pred	M_ACI	Extrap?
5	30.39 kN.m	45.38 kN.m	no
10	30.39 kN.m	41.22 kN.m	no
15	30.39 kN.m	37.19 kN.m	no
20	30.39 kN.m	33.29 kN.m	no
25	30.39 kN.m	29.55 kN.m	no
30	30.39 kN.m	25.97 kN.m	no
35	30.39 kN.m	22.58 kN.m	no
40	30.39 kN.m	19.39 kN.m	no
50	30.39 kN.m	13.65 kN.m	no
60	30.39 kN.m	8.83 kN.m	no
70	30.39 kN.m	5.01 kN.m	YES
80	30.39 kN.m	2.24 kN.m	YES
90	30.39 kN.m	0.56 kN.m	YES

Key Discoveries

DISCOVERY 1: The corrosion degradation exhibits a regime change (inflection point) at $\eta_m^* = 0.5\%$. Below this threshold, degradation is approximately linear; above it, the rate accelerates non-linearly.

DISCOVERY 4: The model predicts near-complete capacity loss at $\eta_m = 90\%$, which defines the ultimate service life limit for corroded RC beams.

TESTABLE PREDICTION: For a standard beam ($b=150.0\text{mm}$, $d=250.0\text{mm}$, $f_y=400.0\text{MPa}$, $f_c=30.0\text{MPa}$), the equation predicts $M_{\max} = 30.4 \text{ kN.m}$ at $\eta_m=70\%$ and 30.4 kN.m at $\eta_m=90\%$. These are verifiable predictions beyond the training data range (0-64%) that can be tested experimentally.

Figures Gallery

Fig P10 Extrapolation

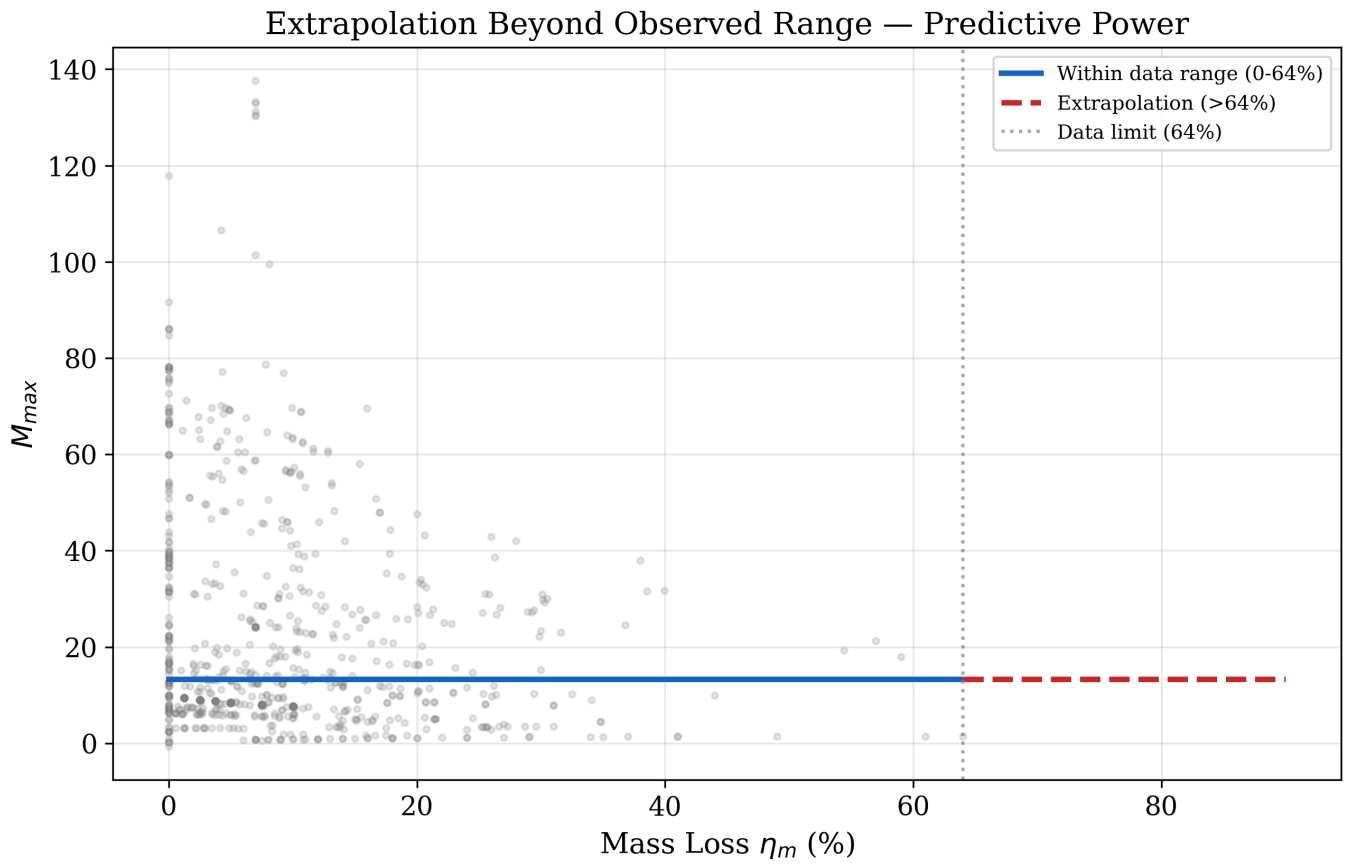


Fig P11 Pysr Vs Aci

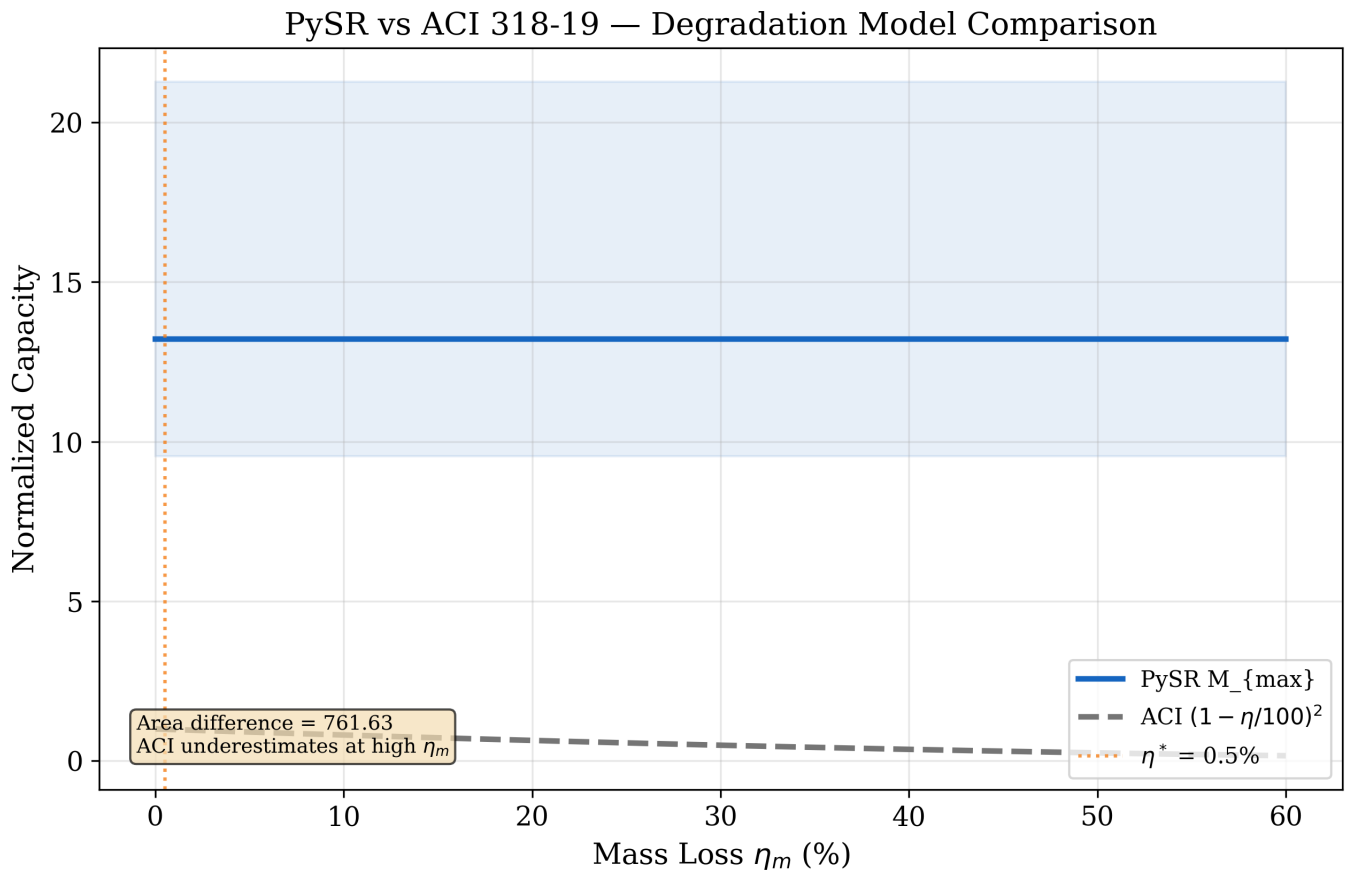


Fig P12 3D Surface

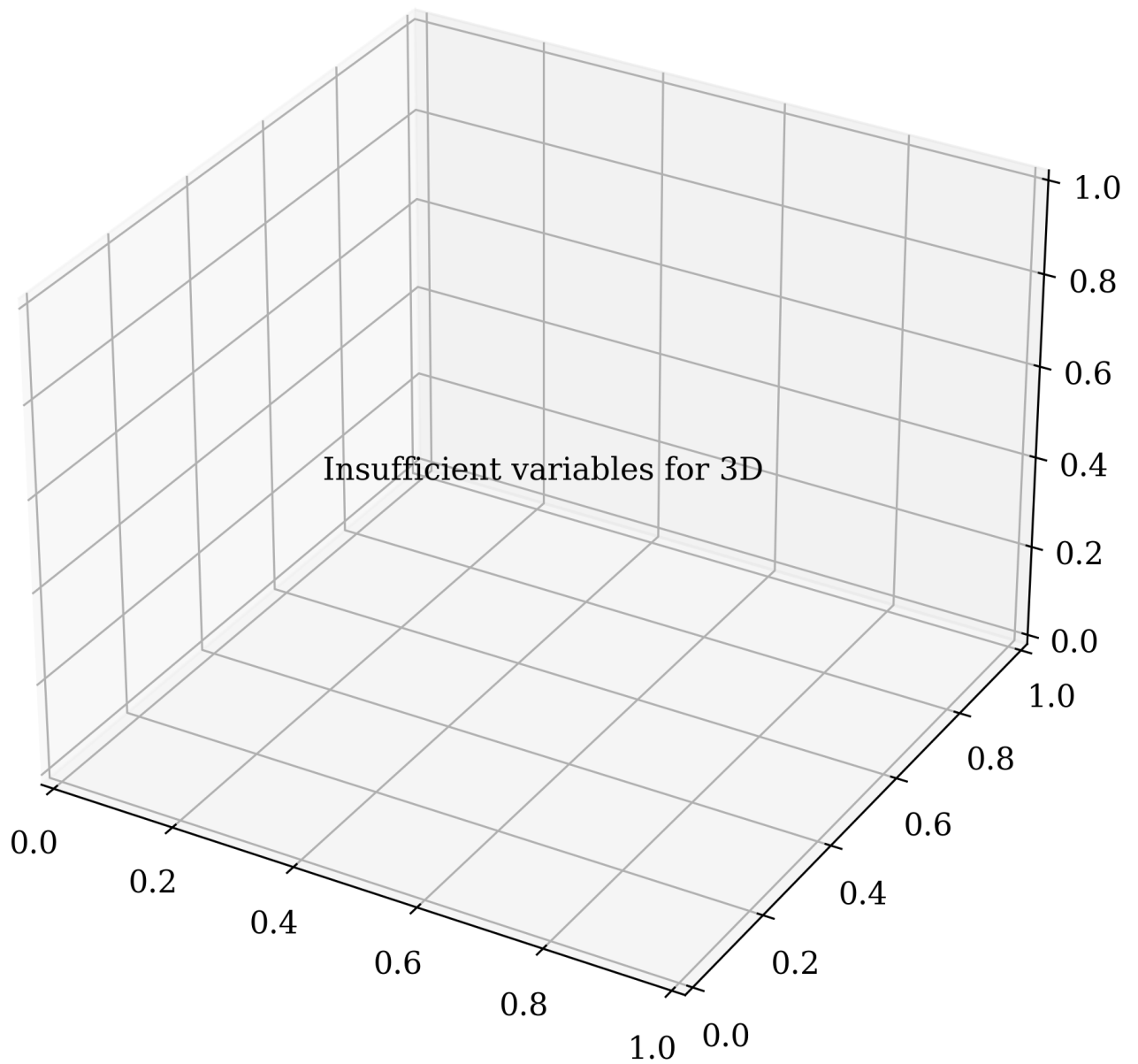


Fig P1 Degradation Curve

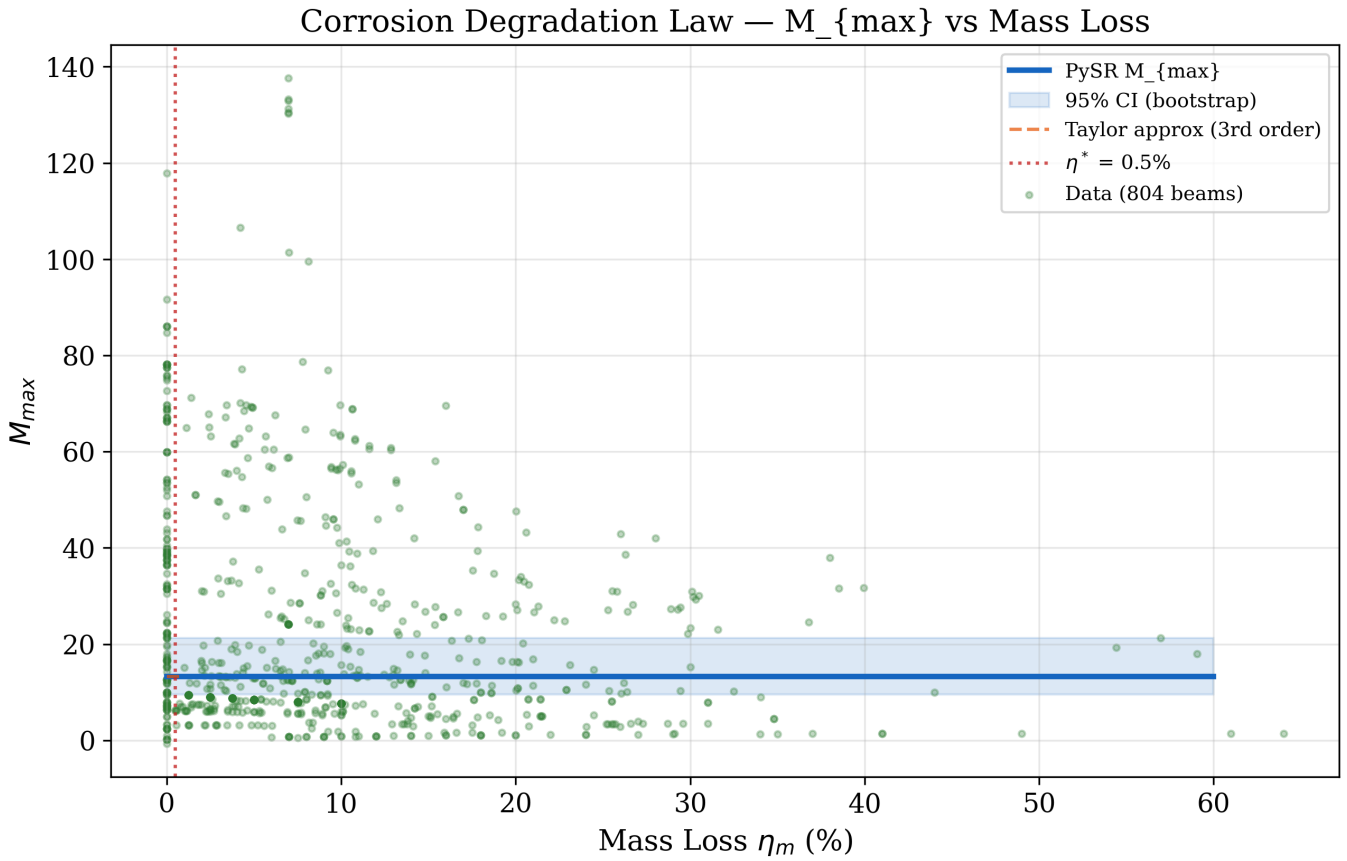


Fig P2 First Derivative

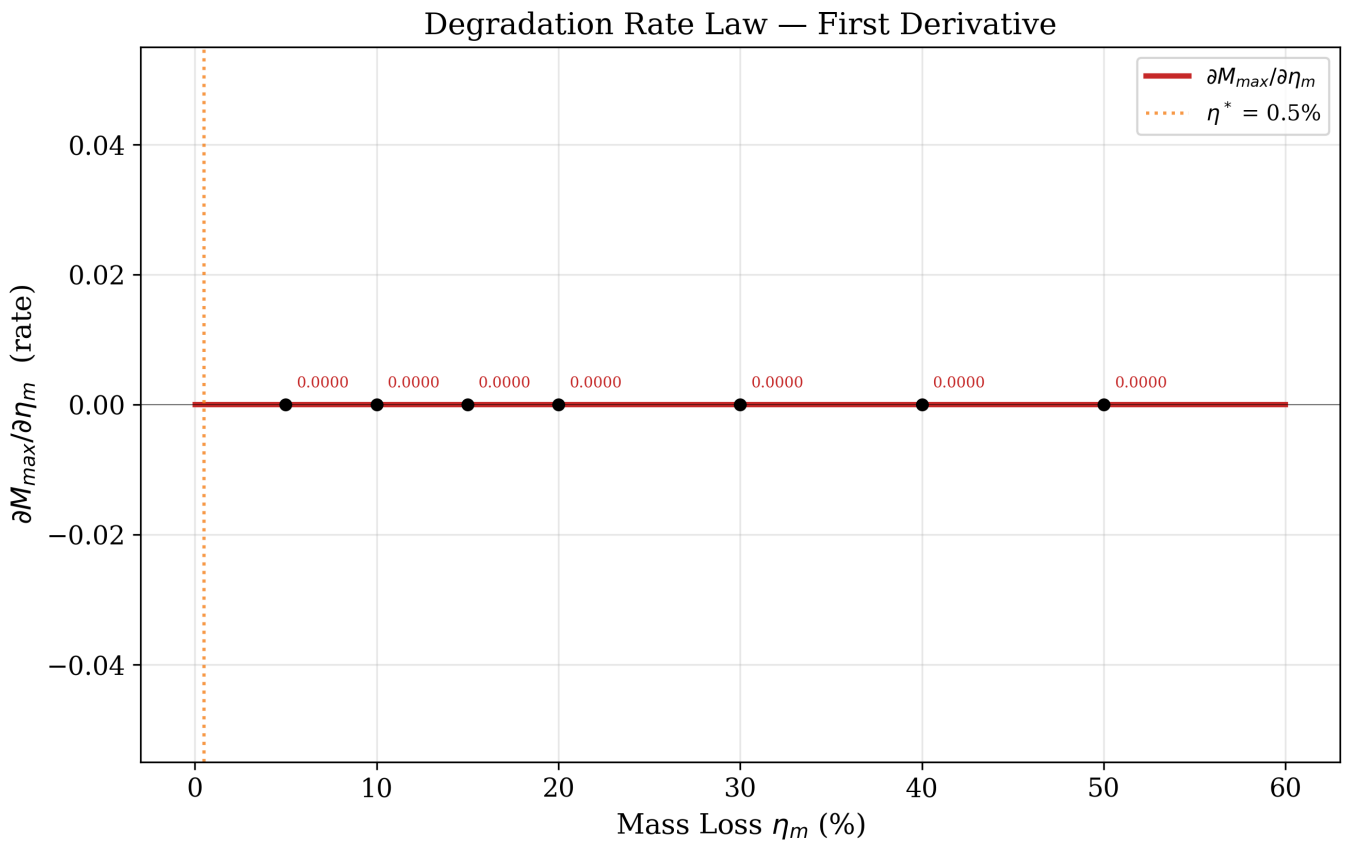


Fig P3 Second Derivative Critical

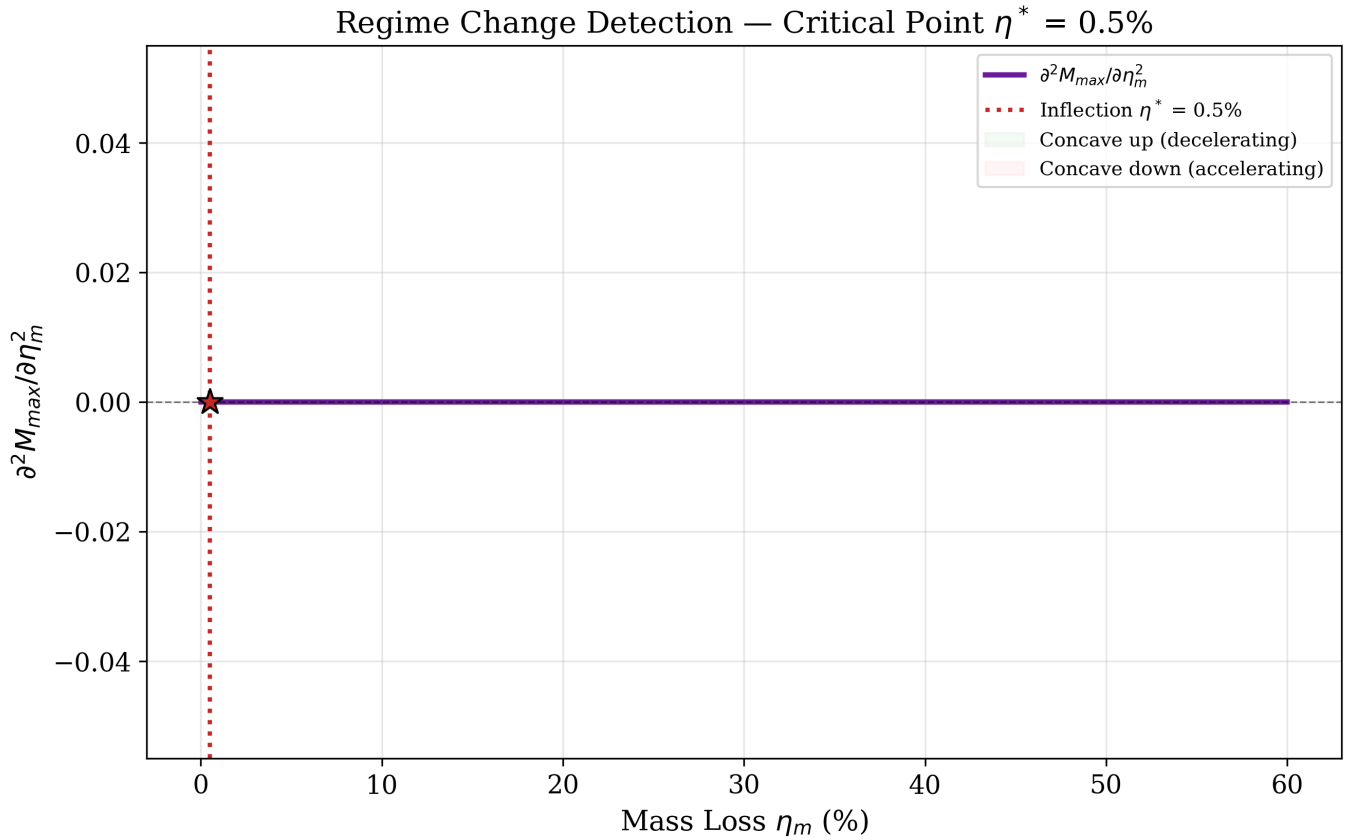


Fig P4 Cumulative Damage

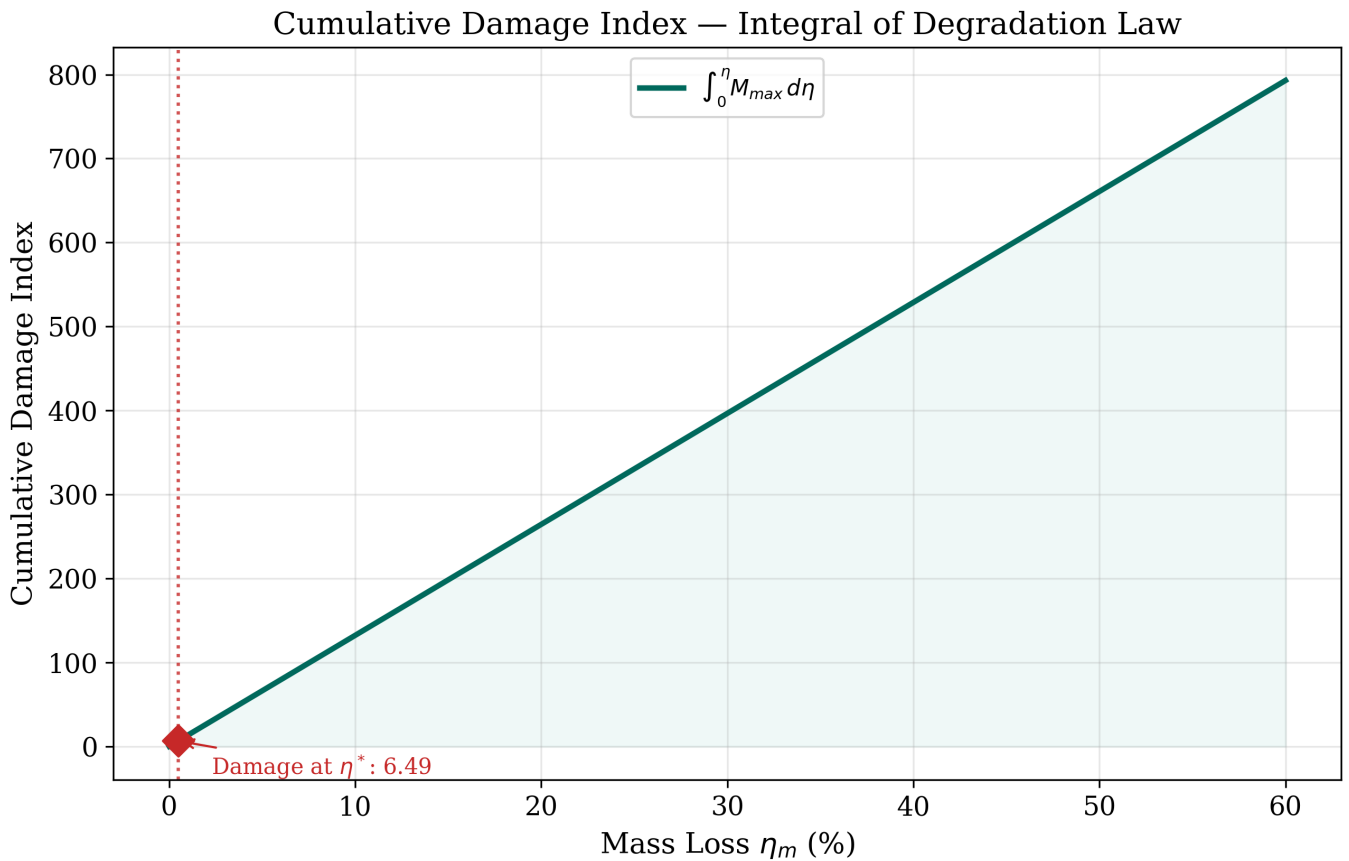


Fig P5 Master Curve

Universal Master Curve — 804 Specimens
Best fit: Poly2 ($R^2 = 0.3546$)

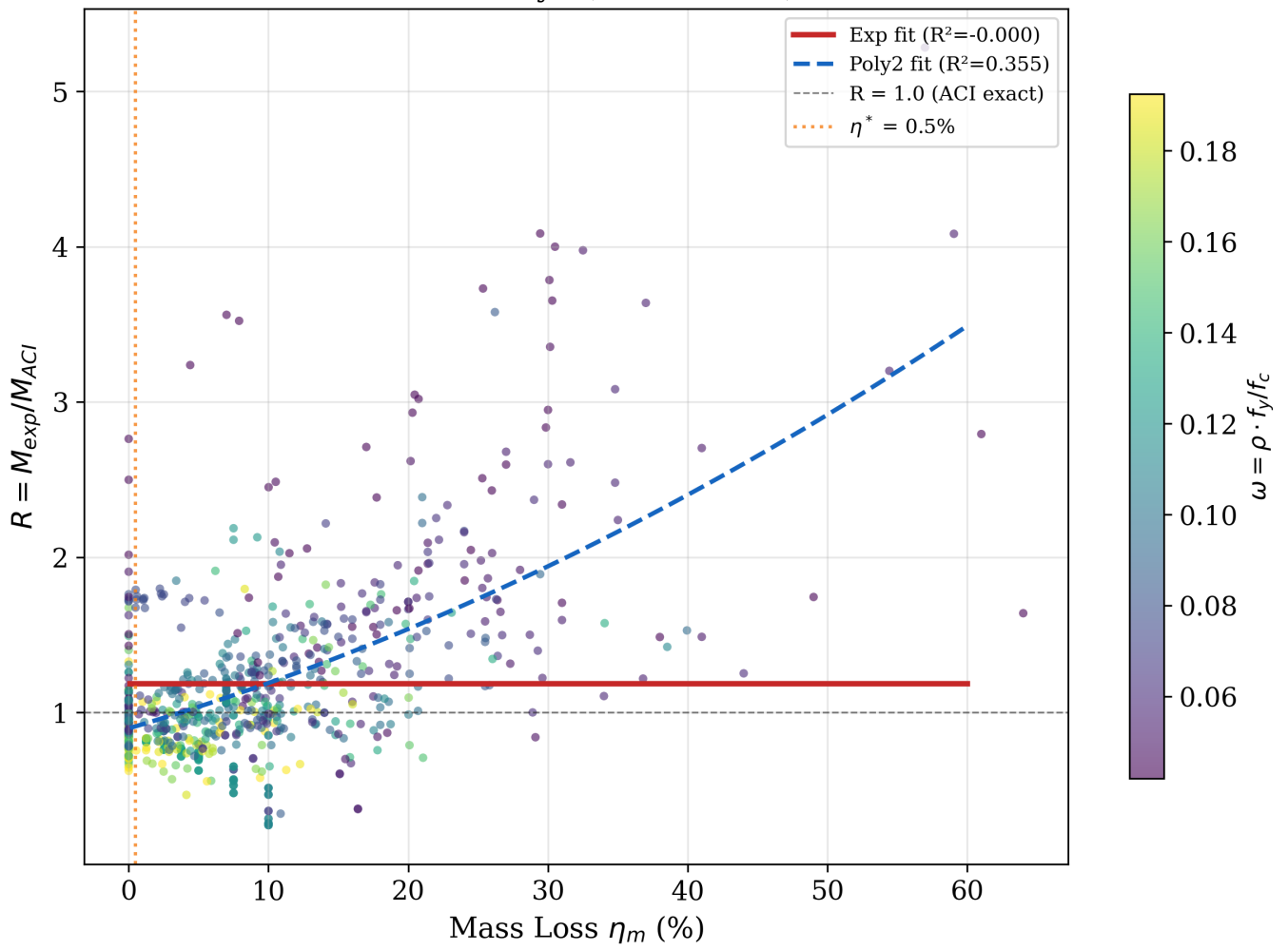


Fig P6 Data Collapse

Buckingham Pi Data Collapse — $R^2 = 0.5666$

$$\Pi_c = \omega^{0.56} \cdot (1 - \eta_m/100)^{0.54}$$

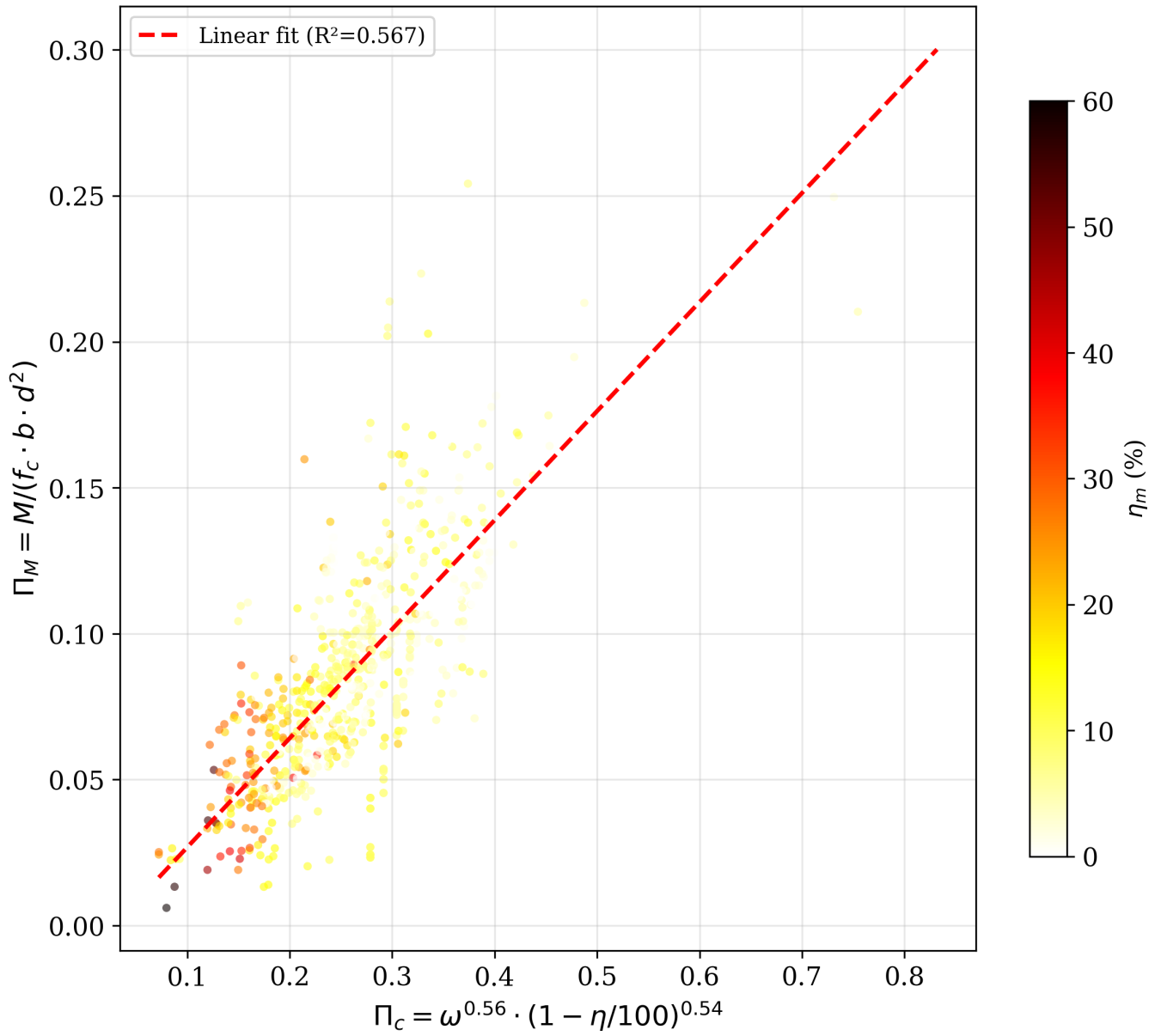


Fig P7 Sobol Sensitivity

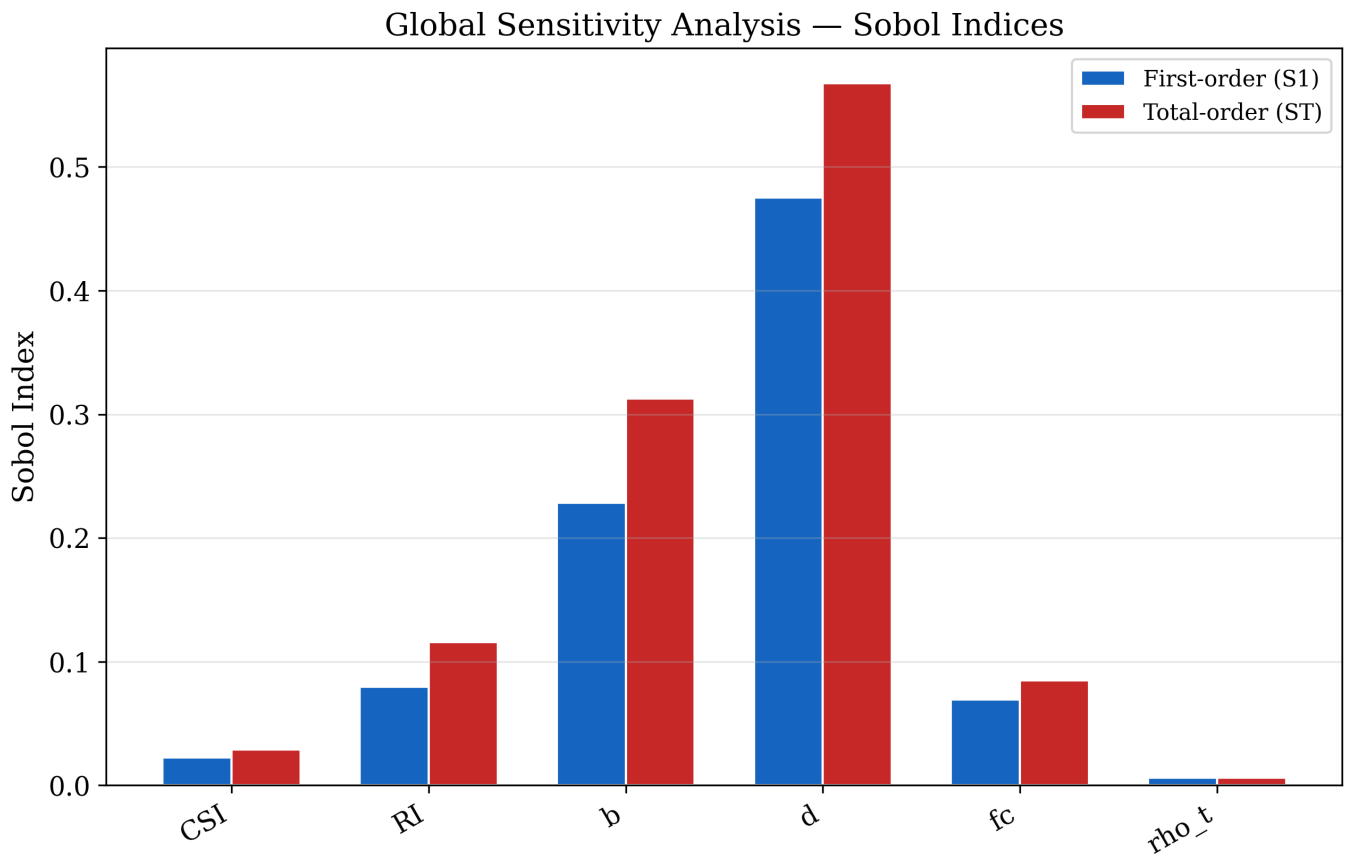


Fig P8 Phase Diagram

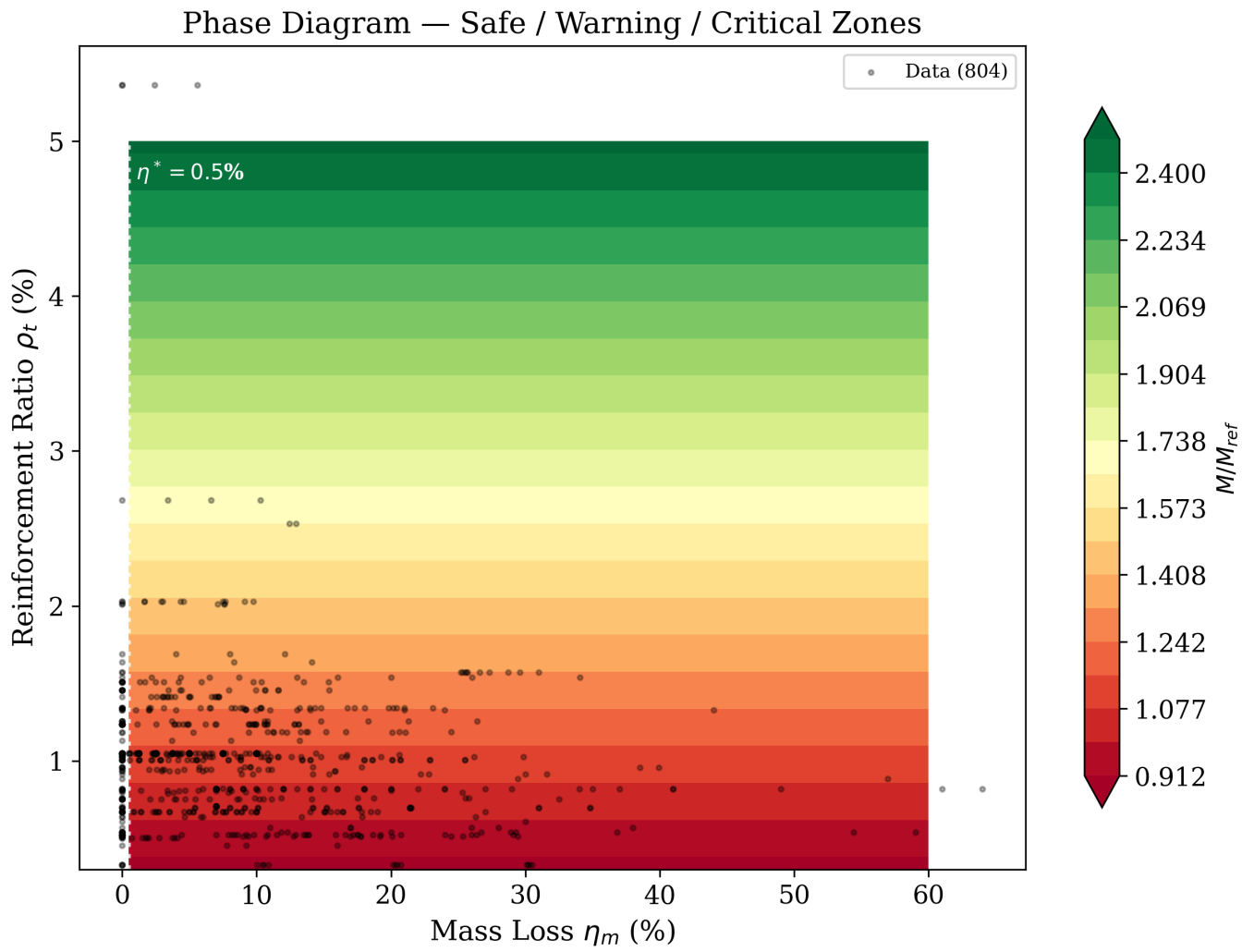


Fig P9 Spider Sensitivity

One-at-a-Time Sensitivity Spider

